

INTELLIGENT MULTI-AGENT URBAN TRAFFIC MANAGEMENT

Advanced Data Science Project

Done by
Bilel Brini, Lina Haddad, Amen Allah Ben Aissa, Hazem Mrad



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01

BACKGROUND & OBJECTIVES

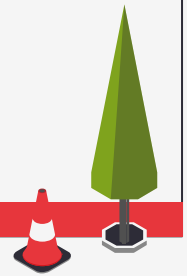
Framing the problem before designing a solution



THE CHALLENGE

Urban traffic congestion is complex and constantly changing driven by time of day, location, demand, and unexpected events. This project explores how Data Science and intelligent autonomous agents can build a smarter, adaptive traffic system, using the SUMO microscopic traffic simulator as the experimentation environment.

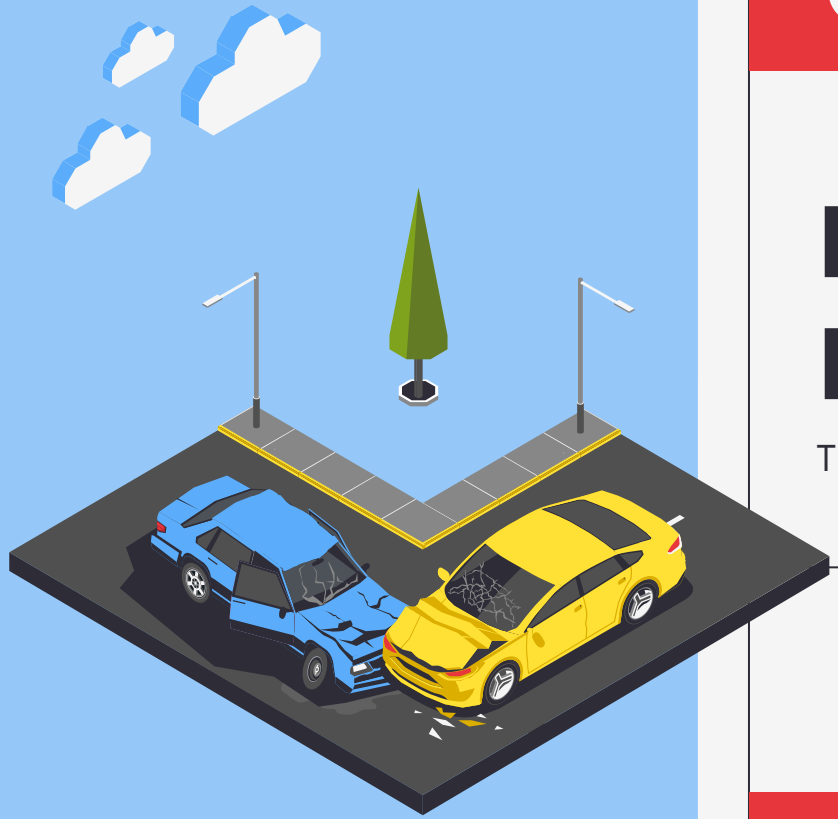
How might we use data and intelligent agents to make urban traffic more efficient, adaptive, and sustainable?" We follow the Design Thinking process Empathize, Define, Ideate, Prototype, Test to move from real stakeholder pain points to a working, tested prototype.



PROJECT OBJECTIVES

- Follow Design Thinking: Empathize, Define, Ideate, Prototype, Test
- Design intelligent agents capable of traffic-control decisions
- Turn raw traffic data into insights and measurable improvements
- Deliver a working prototype, tested against a baseline





02

Emphasize Phase

Three stakeholders, Three very different needs



YOU NEED TO DIVIDE THE CONTENT



MEHDI (Daily Driver)

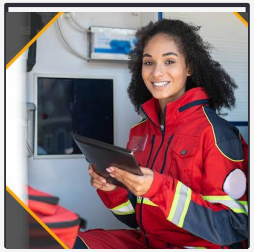
Wants to arrive on time, avoid stops, reduce stress and fuel use frustrated by unadaptive lights and poor emergency coordination



SARRA (Pedestrian)

Wants safe, adaptive crossing times frustrated by long unpredictable waits and drivers ignoring crosswalks

YOU NEED TO DIVIDE THE CONTENT



SONIA (Emergency Responder)

Needs predictable priority passage through intersections frustrated by inconsistent signal response to sirens

Mehdi Ben Yahya | Daily Driver

Persona goal: *Arrive on time, avoid unnecessary stops, anticipate disruptions, and reduce stress, fuel use and pollution.*

ID / scope	Business objective	Data science objective (draft)	Metrics / KPIs
M1 *Core*	Reduce commuters' average trip time and number of stops during peak hours.	Train an adaptive signal-control agent that minimises vehicle delay from real-time intersection state (queue length, waiting time, current phase). *Approach: reinforcement learning (e.g. DQN / PPO) against fixed-time and actuated baselines.*	Mean waiting time and time loss per vehicle Stops per vehicle Throughput (no regression)
M2 *Extension*	Give drivers reliable, accurate travel-time expectations.	Forecast short-term travel time and congestion per road link (5 to 30 min horizon) from detector and vehicle data. *Approach: time-series model (gradient boosting or LSTM) against a historical-average baseline.*	MAE / MAPE of travel time by horizon Travel-time reliability (95th percentile vs mean)
M3 *Extension*	Cut fuel consumption and pollution caused by stop-and-go traffic.	Add fuel / emission terms to the agent's reward and quantify the trade-off between delay and emissions. *Approach: multi-objective reward using SUMO emission outputs (to confirm what SUMO exposes).*	CO_2 and fuel per vehicle-km Idling time Delay vs emissions trade-off curve

Sarra Trabelsi | Pedestrian

Persona goal: *Cross busy roads safely and quickly, with crossing times adapted to how many people are waiting.*

ID / scope	Business objective	Data science objective (draft)	Metrics / KPIs
S1 *Core*	Shorten pedestrian waiting time at busy crossings without hurting vehicle flow.	Extend the agent's state with pedestrian queue counts and add pedestrian waiting time to its reward. ***Approach:*** *pedestrian-aware RL or hybrid rule + learned policy, compared with a fixed-time pedestrian phase.*	Mean and 95th-percentile pedestrian wait Change in vehicle delay and throughput (no-regression constraint)
S2 *Core*	Guarantee enough safe crossing time, scaled to real demand.	Forecast short-term pedestrian arrivals / waiting counts and set the walk-phase duration dynamically. ***Approach:*** *count forecasting plus a duration policy (rule-based first, learned later), tested with SUMO pedestrian flows.*	% of crossings completed before phase end Walk time vs required time Wait / crossing time vs crowd size (should scale)
S3 *Optional (d)*	Improve pedestrian safety where vehicles do not respect crossings.	Estimate conflict risk per crossing from simulated vehicle–pedestrian interactions and rank locations for timing or layout changes. ***Approach:*** *SUMO surrogate safety measures plus a simple risk score.*	Conflict / near-miss events per crossing Stability of the risk ranking across runs

Sonia | Emergency Responder

Persona goal: *Reach emergencies and hospitals as fast as possible, safely, with predictable priority through intersections.*

ID / scope	Business objective	Data science objective (draft)	Metrics / KPIs
E1 *Core*	Reduce emergency response time compared with fixed-timer operation.	Coordinate priority across several intersections so agents create a green corridor for an emergency vehicle along its route. ***Approach:*** *rule-based pre-emption as baseline, then cooperating / communicating agents.*	Emergency vehicle travel time and red-light stops vs baseline (% reduction) Extra delay imposed on civilian traffic
E2 *Core*	Treat emergency vehicles as a distinct urgent class, with consistent signal response.	Detect emergency vehicles and predict their arrival time at each downstream intersection so priority starts early enough. ***Approach:*** *EV type flag in SUMO plus ETA regression on route, speed and queue length.*	ETA error % of intersections already clear on arrival Pre-emption lead time
E3 *Extension*	Give responders route visibility instead of relying on instinct and radio dispatch.	Recommend the fastest route using predicted (not only current) link travel times, with re-routing on incidents. ***Approach:*** *shortest path with forecast edge costs (reuses the M2 forecasting layer).*	Route travel time vs static shortest path Re-route success rate



03

DEFINE PHASE

From insights to a sharp problem statement

CORE PROBLEM STATEMENT

Urban traffic control today is reactive, locally scoped, and car-centric: signals run on fixed patterns, operators lack network-wide visibility, emergency vehicles have no coordinated priority, and pedestrians face crossing times that ignore real demand.

There is a need for an adaptive, network-aware, multi-agent traffic management system that gives operators early visibility, coordinates emergency priority, adapts to pedestrian demand, and proves its improvements with data.



HERE ARE THREE IMPORTANT IDEAS



FASTER RESPONSE

Reduced time-to-detect and time-to-respond to incidents



EMERGENCY PRIORITY

Measurable reduction in emergency response time vs. fixed-timer baseline



PEDESTRIAN WAIT

Reduced average pedestrian wait time, scaling with crowd size

SDGs



SDG 3

Relevant to our **pedestrian safety work**, especially adapting crossing times and making intersections safer.



SDG 7

Relevant for reducing **unnecessary stops, wait times, fuel consumption, and emissions**, as long as we actually measure these in our simulations. Our project already mentions reducing fuel consumption and pollution.

SDGs



SDG 9

Our system is a **smart, flexible, and resilient** way to manage transport infrastructure, one that can respond to changing traffic, incidents, and emergencies



SDG 11

This is one of the targets **most directly tied** to our project: smart transport, road safety, accessibility, pedestrians, and adapting to traffic flow.



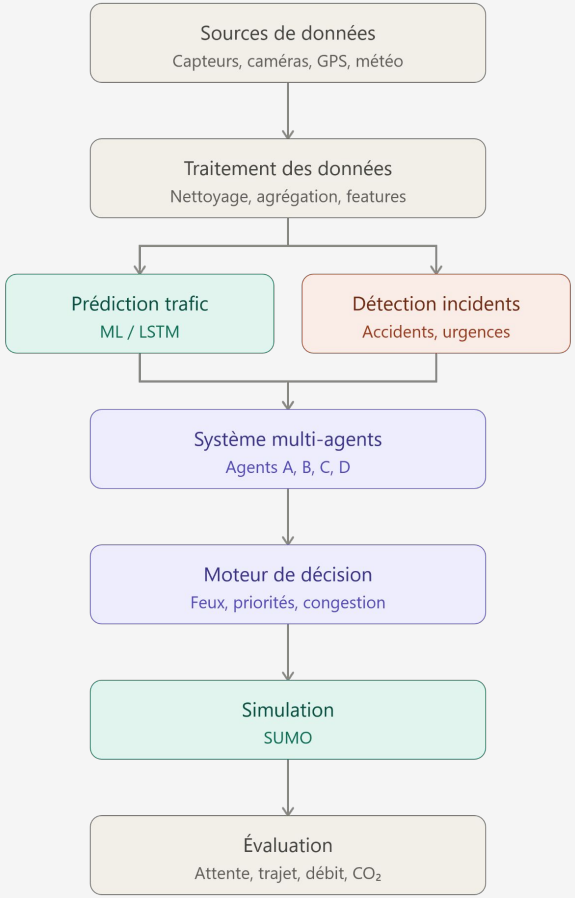
04

SYSTEM ARCHITECTURE

How the agents, SUMO, and dashboard fit together



Architecture



THANKS!

DO YOU HAVE ANY QUESTIONS?



DIVIDE THE CONTENT IN FOUR IDEAS



MARS

Mars is actually a very cold planet



VENUS

Venus has extremely high temperatures



JUPITER

Jupiter is the biggest planet of them all



SATURN

Saturn is a gas giant and has several rings

REVIEWING CONCEPTS IS A GOOD IDEA



MARS

Mars is actually a very cold planet



VENUS

Venus has extremely high temperatures



NEPTUNE

Neptune is very far away from us



MERCURY

Mercury is the closest planet to the Sun



SATURN

Saturn is a gas giant with several rings



JUPITER

Jupiter is the biggest planet of them all

AWESOME WORDS









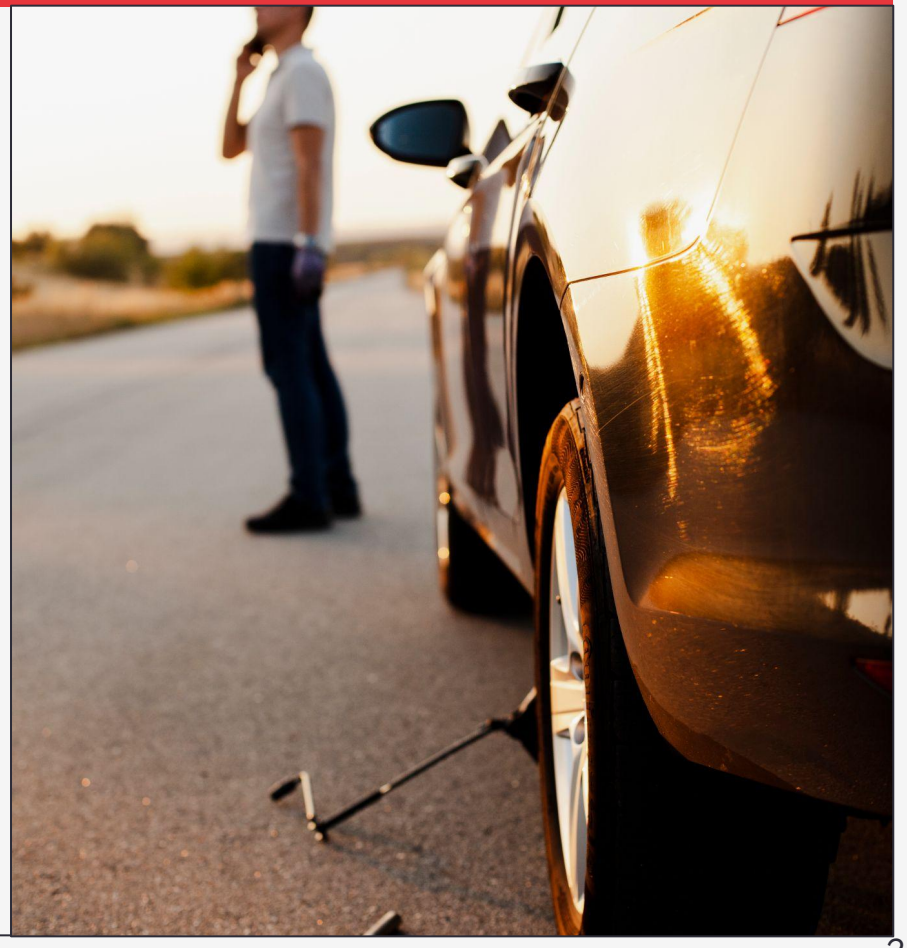
“This is a quote, words full of wisdom that someone important said and can make the reader get inspired.”

—SOMEONE FAMOUS


A PICTURE IS WORTH A THOUSAND WORDS

A PICTURE ALWAYS REINFORCES THE CONCEPT

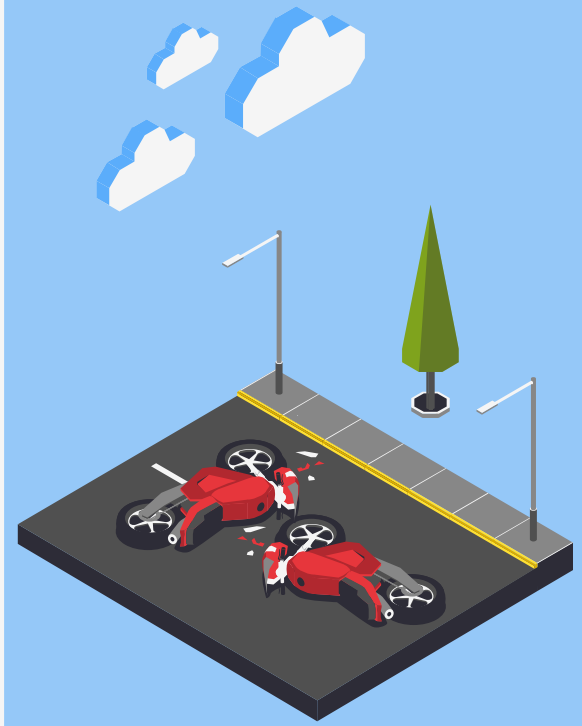
Images reveal large amounts of data, so remember: use an image instead of a long text. Your audience will appreciate it



4,498,300,000

Big numbers catch your audience’s attention







9h 55m 23s

Jupiter’s rotation period

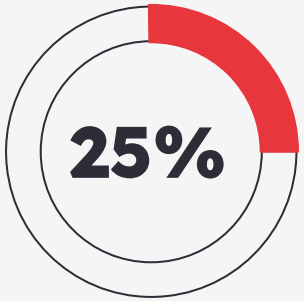
333,000

The Sun’s mass compared to Earth’s

386,000 km

Distance between Earth and the Moon

LET'S USE SOME PERCENTAGES



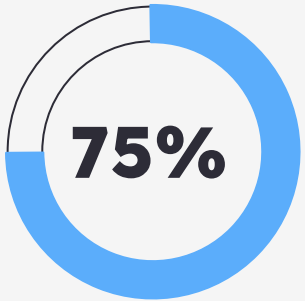
MERCURY

Mercury is the closest planet to the Sun and a very small object



VENUS

Venus has a beautiful name and also a very toxic atmosphere



MARS

Mars is actually a cold planet. It's full of iron oxide dust

COMPUTER MOCKUP

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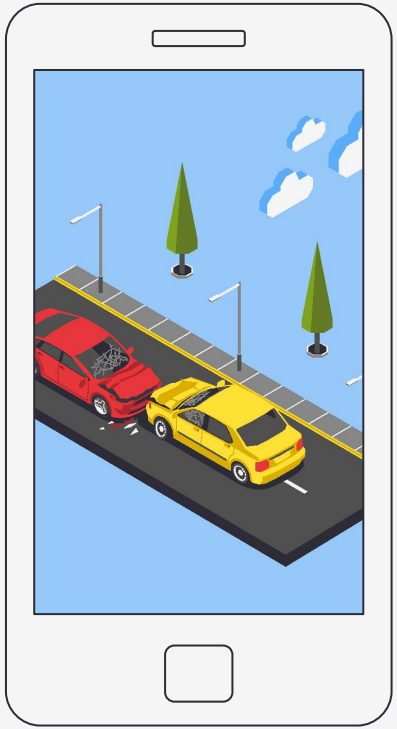


TABLET MOCKUP

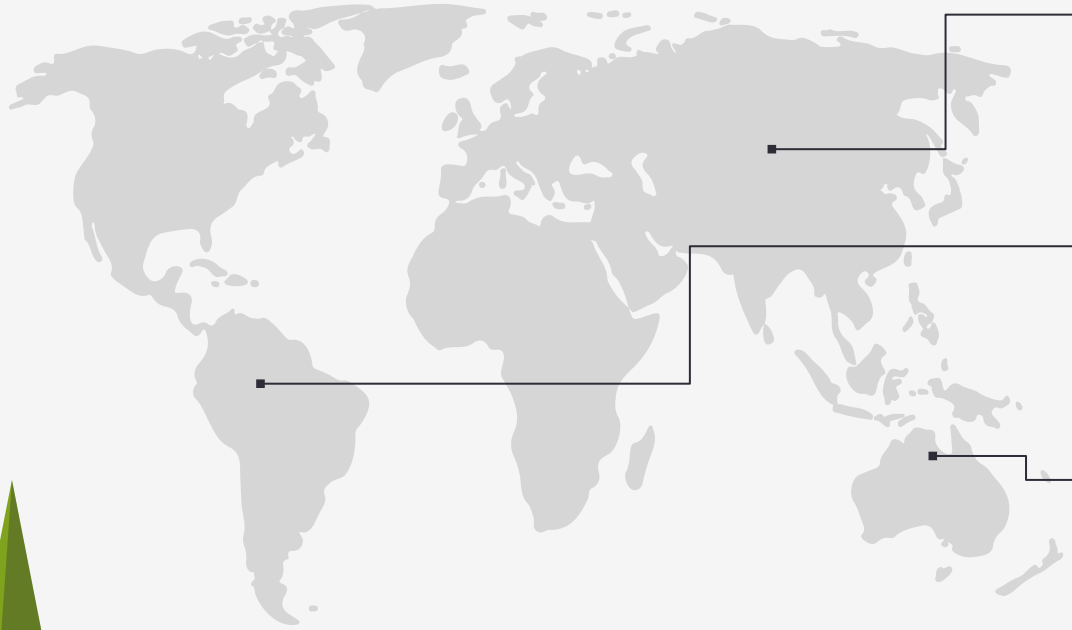
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PHONE MOCKUP

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THIS IS A MAP



MERCURY

Mercury the smallest of them all

VENUS

Venus has extremely high temperatures

SATURN

Saturn is a giant and has several rings

A TIMELINE ALWAYS WORKS WELL

Venus is the second planet from the Sun

VENUS

20XX

Despite being red, Mars is a cold planet

MARS

20XX

20XX

MERCURY

Mercury is the closest planet to the Sun

20XX

JUPITER

Jupiter is the biggest planet of them all

INFOGRAPHICS ARE COOL

MARS

Mars is actually a very cold planet

MERCURY

Mercury the smallest of them all



VENUS

Venus has extremely high temperatures

SATURN

Saturn is a giant and has several rings

TABLES REPRESENT YOUR DATA





	MASS (Earths)	DIAMETER (Earths)	SURFACE GRAVITY (Earths)
MERCURY	0.06	0.38	0.38
MARS	0.11	0.53	0.38
SATURN	95.2	9.4	1.16

YOU CAN USE THIS GRAPH



15%

MERCURY

Mercury is a very small planet

20%

VENUS

Venus has very high temperatures

25%

JUPITER

Jupiter is a gas giant and big

40%

SATURN

Saturn is a gas giant with rings

Follow the link in the graph to modify its data and then paste the new one here. **For more info, click here**

OUR TEAM



SOFIA HILL

You can speak a bit about this person here



KARL HARRIS

You can speak a bit about this person here

THIS IS ANOTHER INFOGRAPHIC



NEPTUNE

Neptune is the farthest planet from the Sun. It's also the fourth-largest planet by diameter in the Solar System



MERCURY

Mercury is the closest planet to the Sun



MARS

Despite being red, Mars is a cold planet



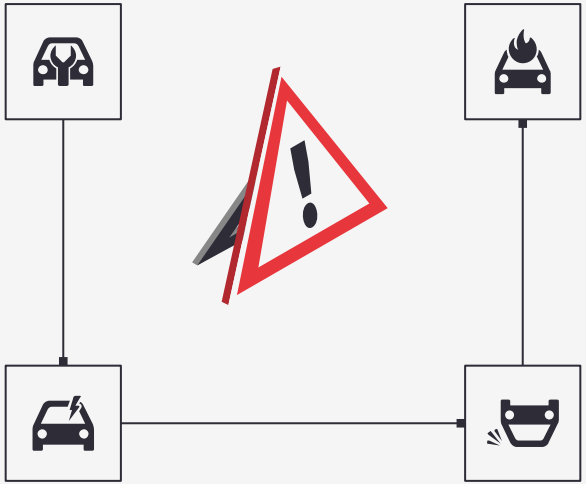
SATURN

Saturn is a giant and has several rings

THIS IS ANOTHER TIMELINE

MARS
Mars is actually a very cold planet

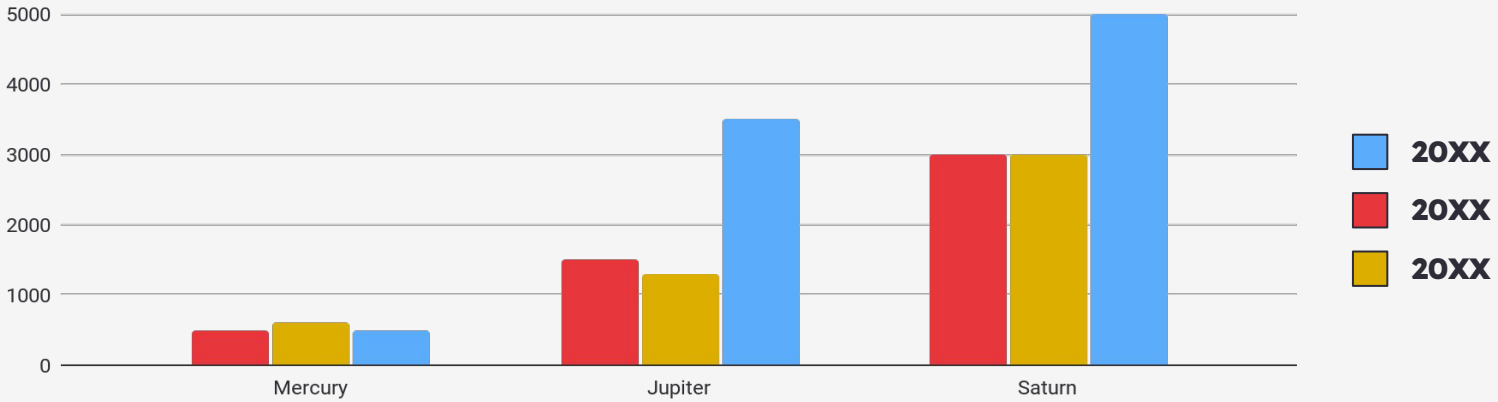
VENUS
Venus has extremely high temperatures



MERCURY
Mercury the smallest of them all

SATURN
Saturn is a giant and has several rings

THIS IS ANOTHER GRAPH



MERCURY

Mercury is a very small planet

JUPITER

Jupiter is a gas giant and big

SATURN

Saturn is a gas giant with rings

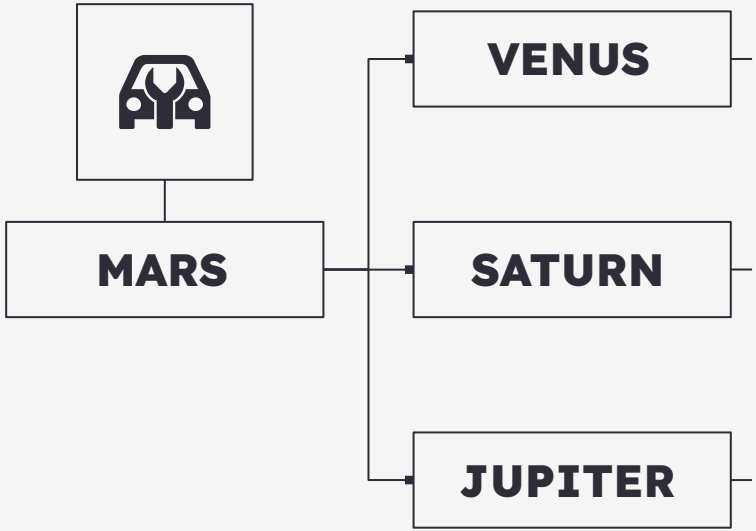
Follow the link in the graph to modify its data and then paste the new one here. **For more info, click here**

THIS IS A ROADMAP

INITIATIVE	OBJECTIVE	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
Neptune is big	Neptune is an ice giant and big												
Earth has life	Earth is the beautiful planet where we live												
Mars is very cold	Despite being red, Mars is a cold planet												
Jupiter is a giant	Jupiter is the biggest planet of them all												
Saturn has rings	Saturn is a gas giant and has several rings												
Mercury is small	Mercury is the closest planet to the Sun												

20XX 20XX 20XX 20XX

THIS IS A TREE DIAGRAM

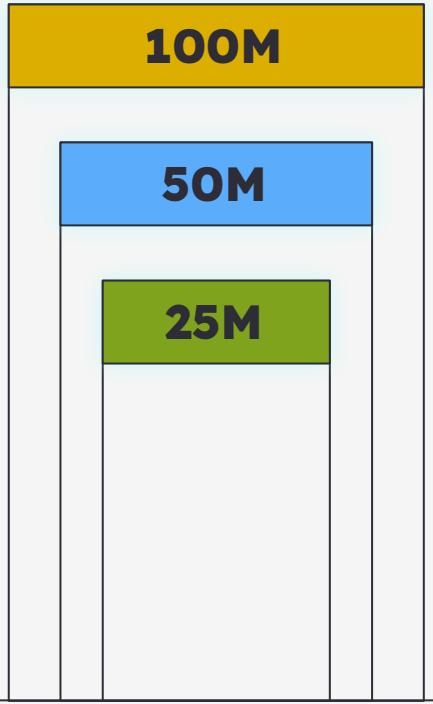


- Venus is the second planet from the Sun
- Venus has extremely high temperatures
- Venus has a toxic atmosphere

- Saturn is a gas giant with rings
- Saturn is made of hydrogen and helium
- Saturn has beautiful rings

- Jupiter is the biggest planet of them all
- Jupiter was named after a Roman god
- Jupiter is a gas giant

THIS IS ANOTHER DIAGRAM



JUPITER

Jupiter is a gas giant and the biggest planet in the Solar System

VENUS

Venus has a beautiful name and is the second planet from the Sun

EARTH

Earth is the third planet from the Sun and the only one that harbors life

THIS IS A TABLE

	LOCATION 1	LOCATION 2	LOCATION 3	LOCATION 4	LOCATION 5
MERCURY	2,500	500	2,750	150	2,750
MARS	150	150	800	2,500	4,500
SATURN	4,500	650	1,200	500	500
VENUS	500	800	6,500	4,500	4,500
JUPITER	2,750	2,750	2,500	800	6,500
EARTH	1,200	4,500	500	1,000	800
MOON	650	1,000	150	650	2,500

ICON PACK I



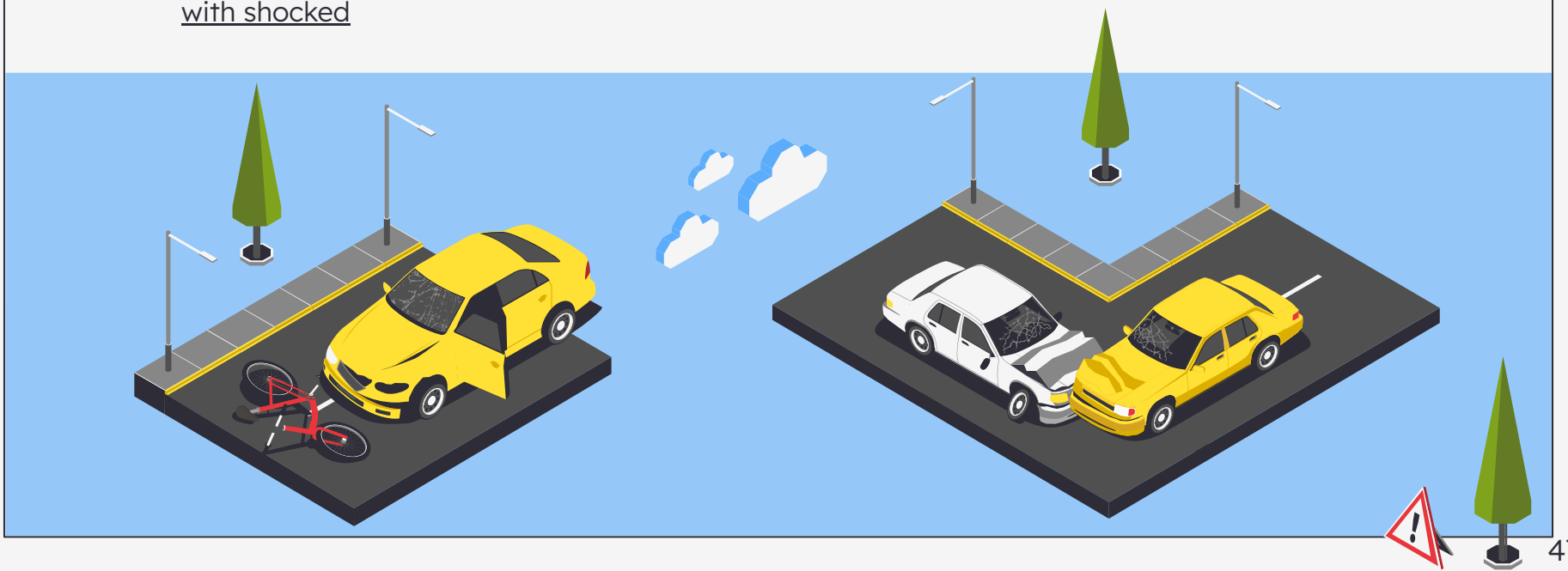
ICON PACK II



ALTERNATIVE RESOURCES

Here’s an assortment of alternative resources whose style fits the one of this template:

- Traffic accident scene isometric composition
- Insurance isometric composition with road intersection outdoor scenery and collision of cars with shocked



RESOURCES

Did you like the resources in this template? Get them for free at our other websites:

PHOTOS

- Man talking on the phone about a car problem
- Young man checking the car engine
- Handsome adult male posing
- Front view of beautiful woman with copy space

ICONS

- Icon Pack: Car accidents
- Icon Pack: Motorway

VECTORS

- Traffic accident isometric composition with crash between car and motorcycle injured motorcyclist on road
- Public city transport isometric composition with two broken cars after collision with ambulance and police car
- Car ownership usage isometric composition with view of two crashed cars with characters of automobile
- Emergency medical care isometric composition



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Pana



Amico



Bro



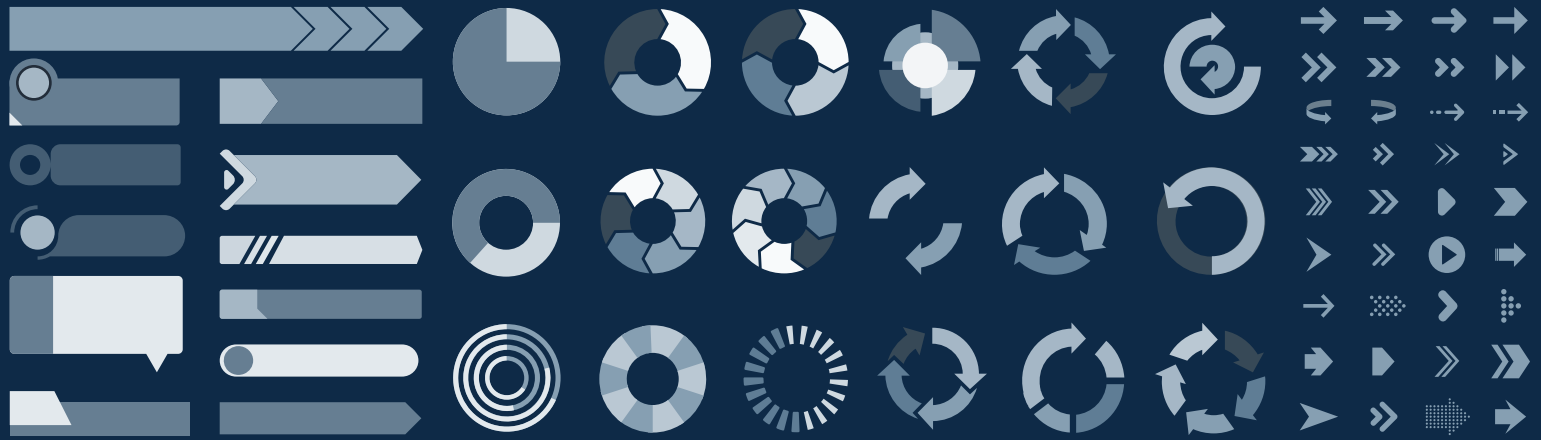
Rafiki



Cuate

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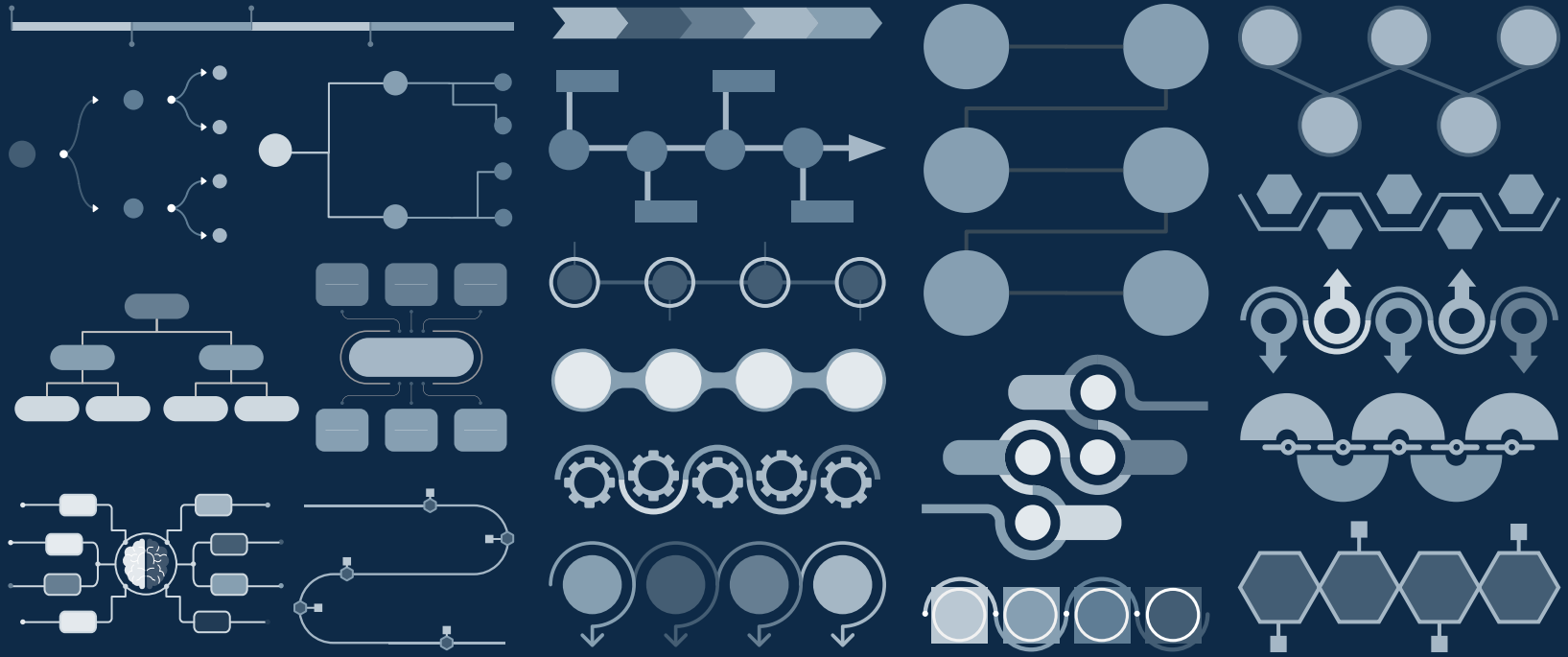


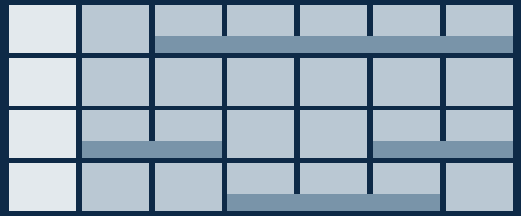


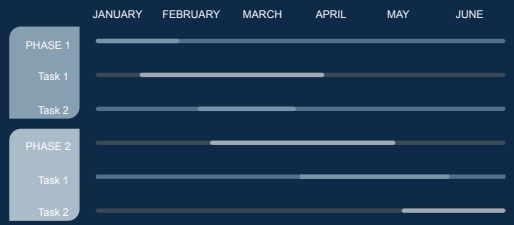


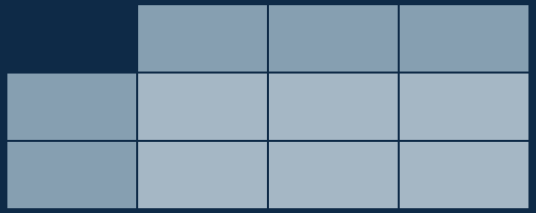


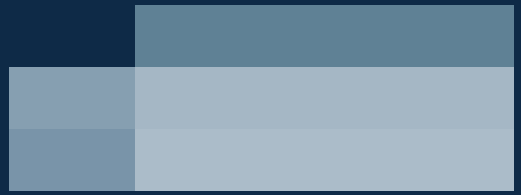


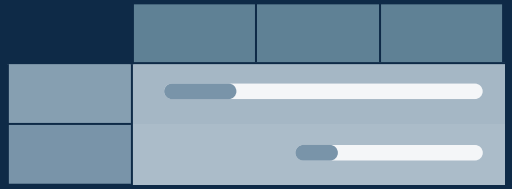


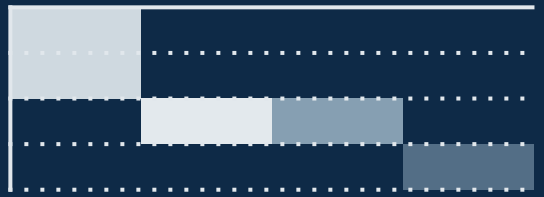



JANUARY
FEBRUARY
MARCH
APRIL
MAY
JUNE
PHASE 1
Task 1
Task 2
PHASE 2
Task 1
Task 2



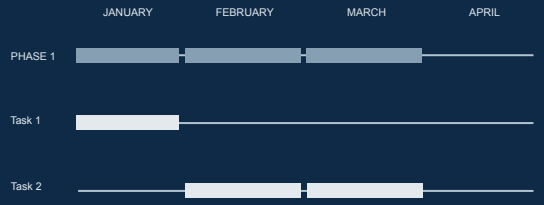


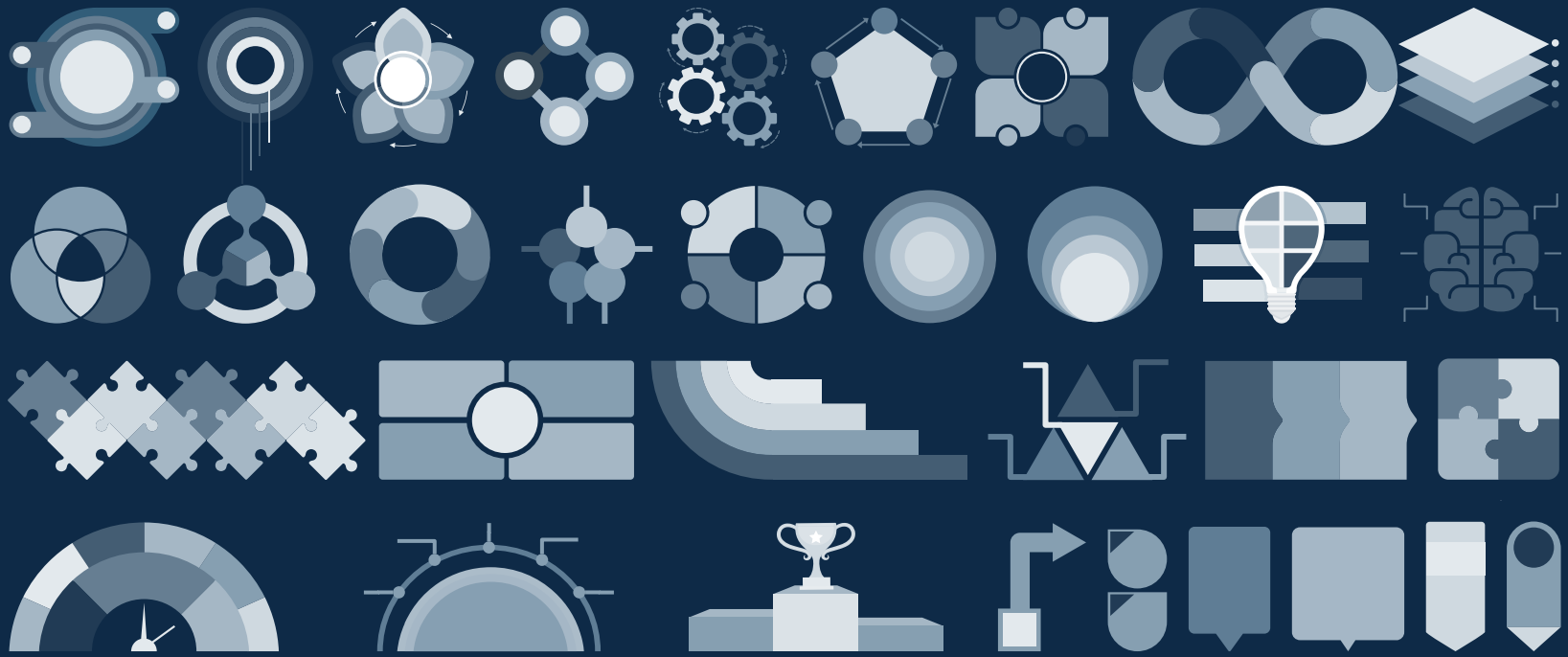


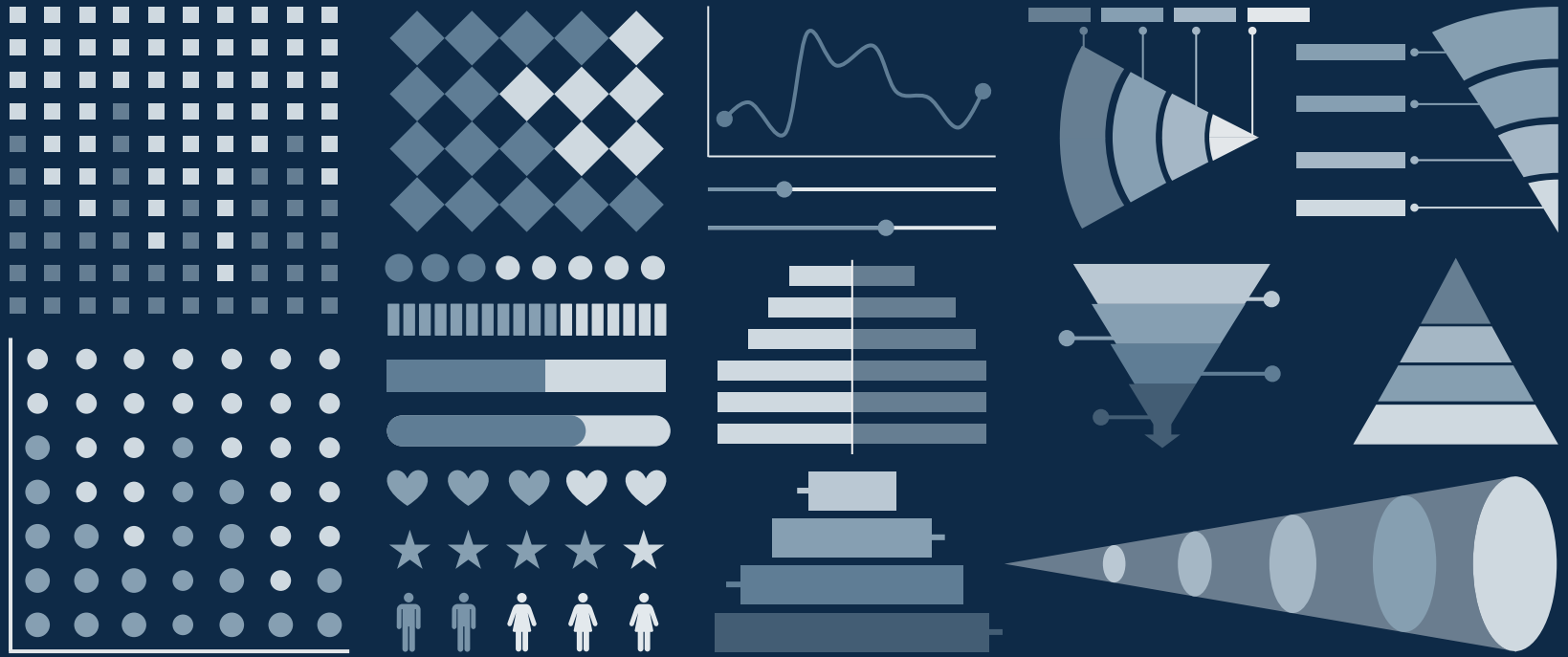







JANUARY
FEBRUARY
MARCH
APRIL
PHASE 1
Task 1
Task 2





...and our sets of editable icons

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In Google Slides, you can also use **Flaticon's extension**, allowing you to customize and add even more icons.



Educational Icons



Medical Icons



Business Icons



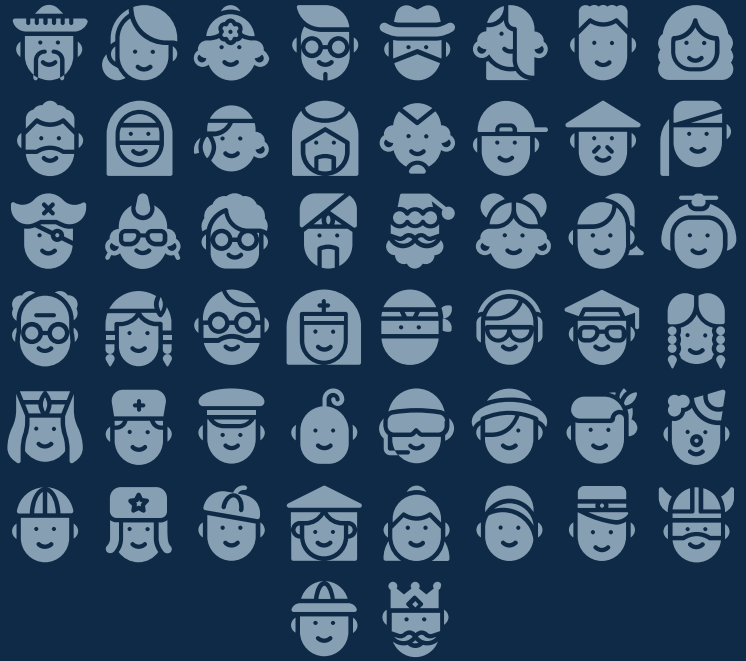
Teamwork Icons



Help & Support Icons



Avatar Icons



Creative Process Icons



Performing Arts Icons



Nature Icons



SEO & Marketing Icons




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